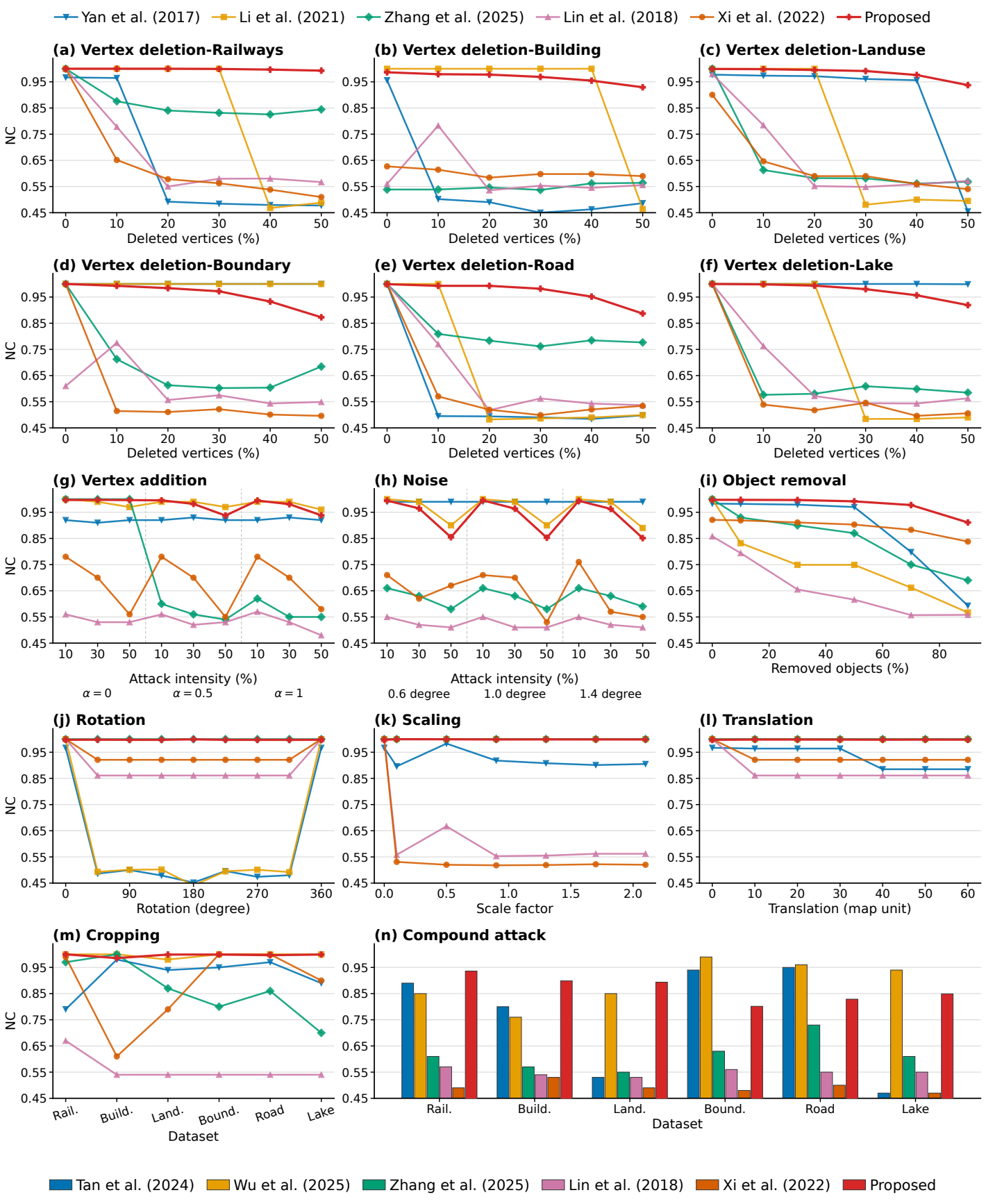

Yan et al. (2017)
Li et al. (2021)
Zhang et al. (2025)
Lin et al. (2018)
Xi et al. (2022)
Proposed
(a) Vertex deletion-Railways
(b) Vertex deletion-Building
(c) Vertex deletion-Landuse
(d) Vertex deletion-Boundary
(e) Vertex deletion-Road
(f) Vertex deletion-Lake
(g) Vertex addition
(h) Noise
(i) Object removal
(j) Rotation
(k) Scaling
(l) Translation
(m) Cropping
(n) Compound attack
NC
Deleted vertices (%)
Attack intensity (%)
α = 0
α = 0.5
α = 1
0.6 degree
1.0 degree
1.4 degree
Removed objects (%)
Rotation (degree)
Scale factor
Translation (map unit)
Rail.
Build.
Land.
Bound.
Road
Lake
Dataset
Tan et al. (2024)
Wu et al. (2025)
Zhang et al. (2025)
Lin et al. (2018)
Xi et al. (2022)
Proposed